

UCS503 - Software Engineering

LedgerX

A Decentralized Web3 Security Protocol for Protected Digital Asset Management
and Real-Time Transaction Anomaly Detection

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Introduction

1.1 Background and Context

The rapid growth of blockchain technology and decentralized applications has resulted in an increasing number of users managing cryptocurrency and other digital assets through blockchain-based wallets and smart contracts. Blockchain provides transparency, immutability, and decentralized control; however, these properties alone do not guarantee protection against malicious transactions, rug pulls, abnormal wallet activity, or other security threats.

Traditional cryptocurrency wallets primarily provide mechanisms for storing, receiving, and transferring digital assets. The responsibility of identifying suspicious transactions and responding to potential threats is generally left to the user. This can create a significant security risk because malicious transactions can occur rapidly and may result in irreversible financial losses.

Smart contracts provide an opportunity to automate security mechanisms directly on the blockchain. By encoding predefined rules into smart contracts, a system can automatically perform security-sensitive operations without depending entirely on manual user intervention.

Machine learning can further improve this process by analysing transaction characteristics and identifying behaviour that deviates from normal transaction patterns. Combining blockchain-based automation with anomaly detection therefore provides an opportunity to create a more proactive approach to digital asset security.

LedgerX, also referred to as the SaveMe Protocol, is designed as a decentralized Web3 security protocol that combines smart-contract automation with transaction anomaly detection. The system provides a protected wallet environment in which transactions can be monitored and potentially suspicious activity can trigger an automated protection mechanism.

The current web application provides users with an interface for connecting a cryptocurrency wallet and accessing the major LedgerX functions. These include sending assets, viewing transactions, working with tokens, creating protocol-related assets, and viewing the application's monitoring interface.

1.1.1 Problem Statement

Cryptocurrency users are exposed to several security threats, including malicious transactions, abnormal wallet activity, and rug pulls. Once a transaction has been executed on a blockchain, it is generally difficult or impossible to reverse it through conventional means.

Existing wallet systems primarily provide transaction functionality but may require users to manually identify suspicious activity and take appropriate action. This creates a gap between the detection of a potential threat and the execution of a protective response.

The problem addressed by LedgerX is therefore the lack of an integrated system that can monitor blockchain transactions, identify potentially anomalous activity, and automatically initiate a predefined protection mechanism to reduce the risk of digital asset loss.

LedgerX addresses this problem by combining a protected smart-contract-based wallet with transaction monitoring, anomaly detection, liquidity conversion, and an automated asset-protection workflow.

1.2 Project Objectives

The major objectives of LedgerX are:

1. **Protected Wallet Creation:** Develop a smart-contract-based protected wallet that allows users to store and manage digital assets within an environment equipped with automated security mechanisms.
2. **Transaction Anomaly Detection:** Implement a mechanism for analysing blockchain transactions and identifying potentially suspicious or abnormal activity.
3. **Automated Threat Protection:** Develop an automated protection mechanism that can respond when suspicious activity is detected and initiate the appropriate security operations.
4. **Liquidity Conversion and Fund Recovery:** Provide a mechanism for converting assets into a safer or more stable form and recovering protected funds for the asset owner when a threat is detected.
5. **Transaction Monitoring:** Provide a user-friendly interface through which users can monitor wallet activity and inspect transaction history.

These objectives are Specific because each objective corresponds to a distinct capability of the LedgerX protocol. They are Measurable through functional implementation and system-level evaluation. They are Achievable using blockchain, smart-contract, web development, and machine-learning technologies. They are Relevant to the problem of improving digital-asset security and are Time-bound to the development and evaluation milestones of the semester project.

Methodology and Design

2.1 System Architecture

LedgerX follows a modular architecture combining a web-based frontend, blockchain smart contracts, transaction monitoring, and an anomaly-detection component.

The frontend acts as the primary interaction layer through which users connect their cryptocurrency wallets and interact with the protocol. The blockchain layer contains the smart contracts responsible for wallet management, asset transfers, liquidity management, and security-related operations.

The anomaly-detection component analyses transaction characteristics and identifies potentially suspicious activity. The result of this analysis can be used to initiate the protection workflow.

The major components of the system are:

1. **User Interface:** Provides the web-based interface for connecting wallets, sending assets, viewing transactions, working with tokens, and accessing the core LedgerX functionality.
2. **Web3 Integration Layer:** Connects the frontend application to cryptocurrency wallets and deployed blockchain smart contracts.
3. **Smart Contract Layer:** Implements protected-wallet functionality, asset management, liquidity operations, and automated protection mechanisms.
4. **Anomaly Detection Layer:** Analyses transaction characteristics and identifies potentially abnormal or suspicious transactions.
5. **Liquidity Layer:** Provides the mechanism required for converting assets into a safer form during the protection workflow.

2.1.1 High-Level Design

The high-level design of LedgerX is represented through the architectural and UML diagrams included in this chapter. The diagrams describe the functional interactions, process flow, component communication, data movement, and structural relationships of the system.

Use Case Diagram

The Use Case Diagram identifies the main actors and the functions available in the LedgerX system. The wallet owner interacts with the system to connect a wallet, create a protected wallet, transfer funds, monitor wallet activity, and view transaction information. The security workflow includes anomaly detection, triggering the self-destruct mechanism, token swapping, and returning protected funds to the owner.

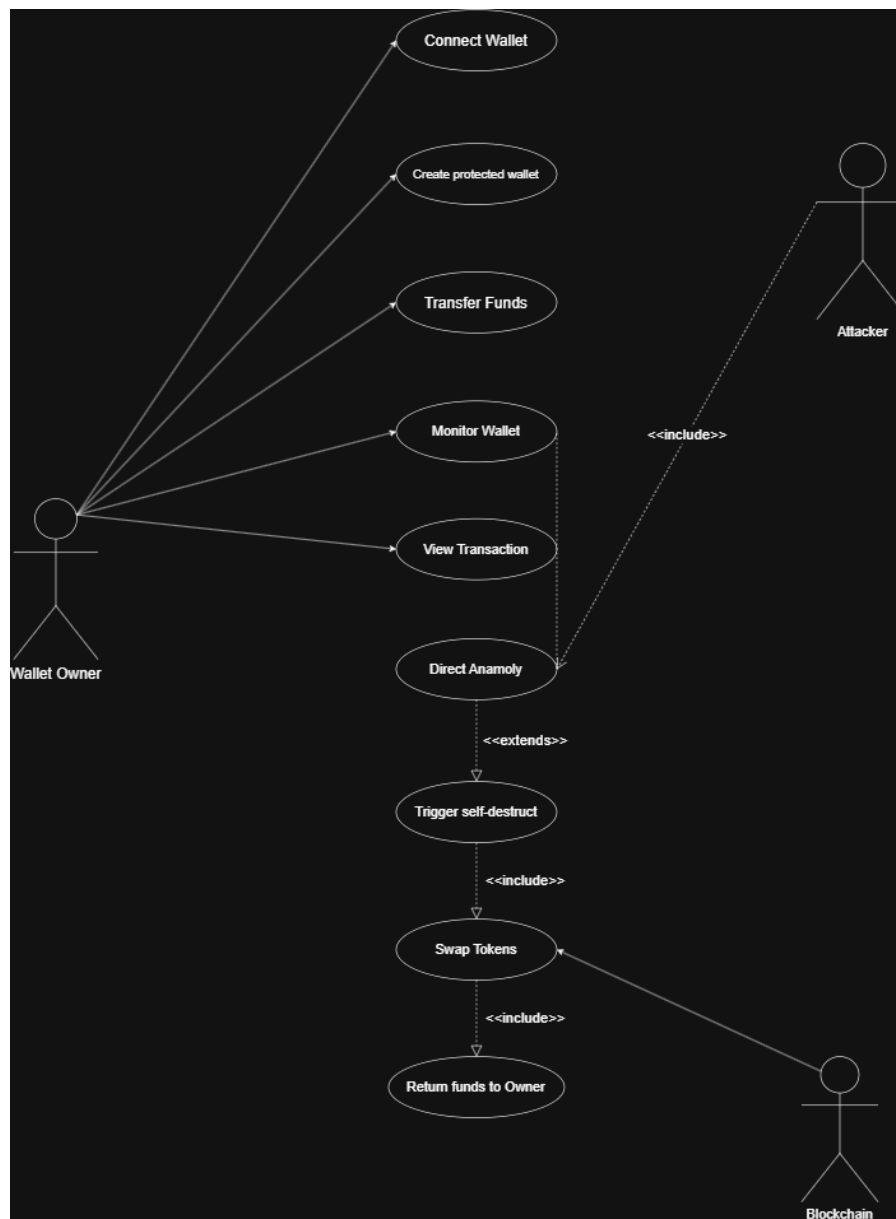


Figure 2.1: Use Case Diagram of LedgerX

Activity Diagram

The Activity Diagram represents the main workflow followed by a LedgerX user. The process begins with wallet connection and protected wallet creation, followed by transfer of funds and transaction monitoring. If no threat is detected, monitoring continues. If a threat is detected, the protection mechanism is triggered, assets are swapped through the liquidity pool, and the recovered funds are returned to the owner.

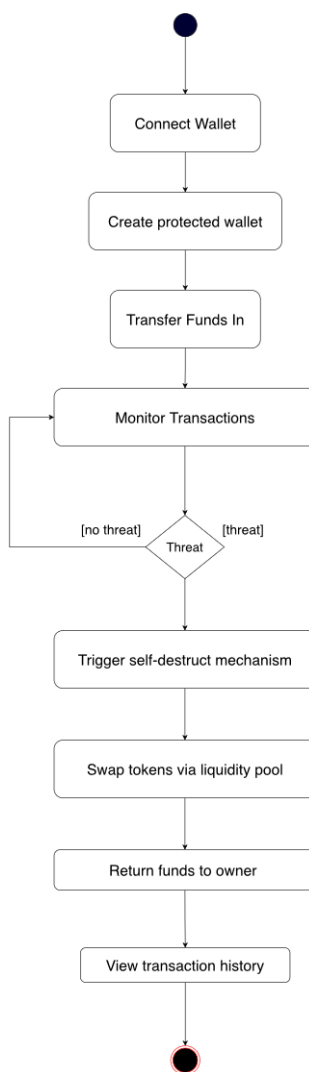


Figure 2.2: Activity Diagram of LedgerX

Swimlane Activity Diagram

The Swimlane Diagram provides another representation of the same core workflow by separating responsibilities between the User, System (Anomaly Guard), and Liquidity Pool. This representation makes the responsibility of each subsystem explicit and shows how control moves between the components during the protection process.

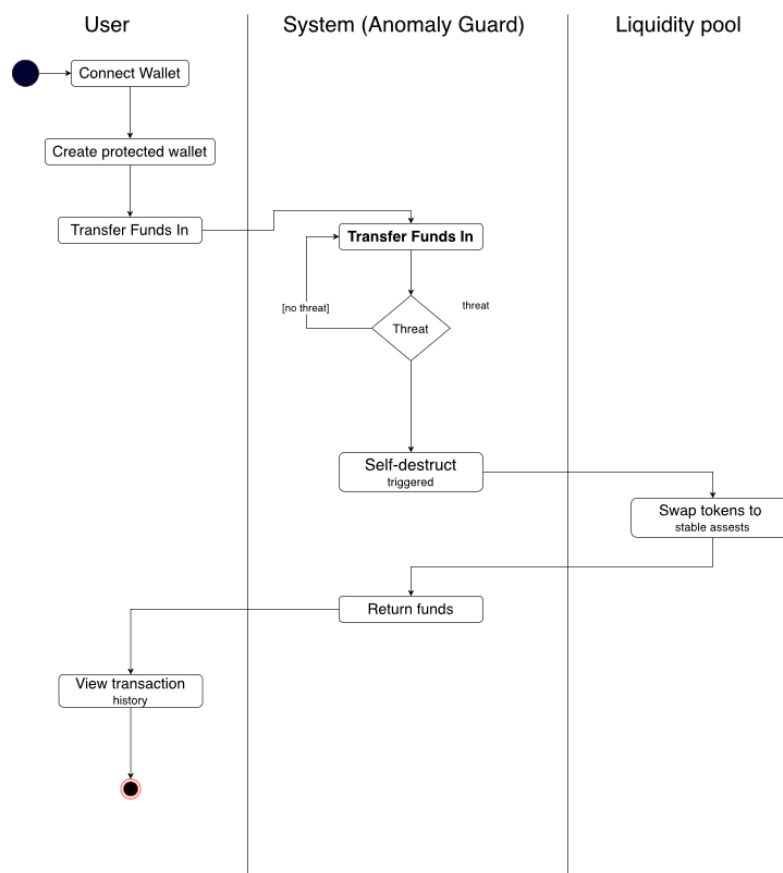


Figure 2.3: Swimlane Activity Diagram of LedgerX

Sequence Diagram

The Sequence Diagram describes the chronological interaction between the User, Protected Wallet, Anomaly Guard, and Liquidity Pool. It shows wallet connection and creation, transfer of funds, transaction monitoring, anomaly detection, activation of the protection mechanism, asset swapping, and return of recovered funds.

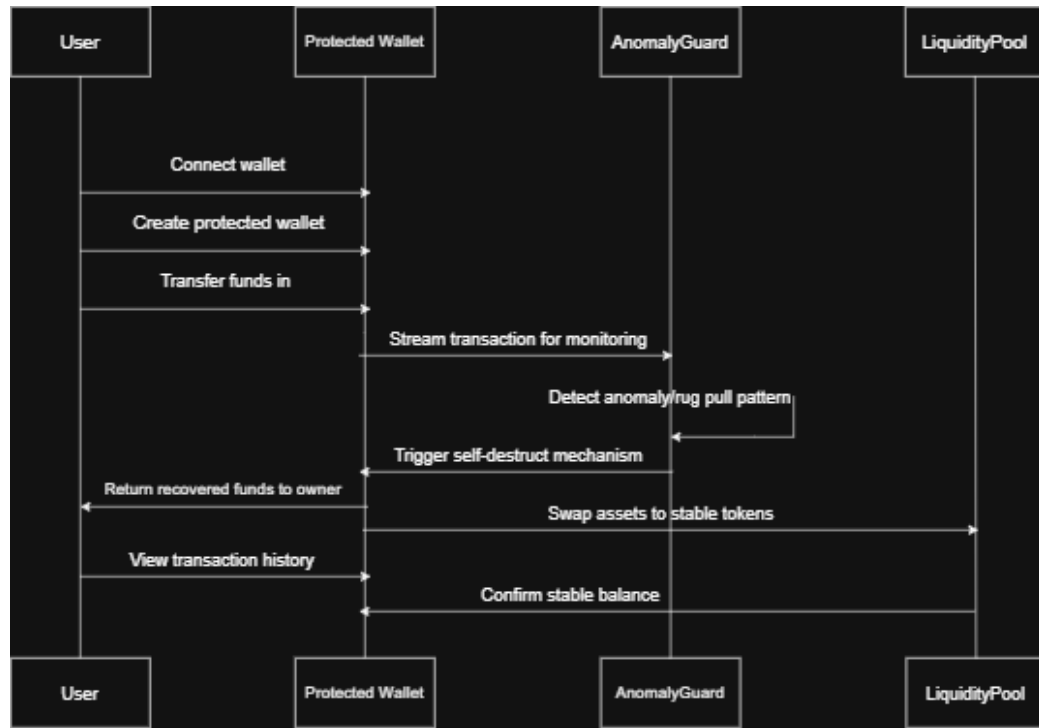


Figure 2.4: Sequence Diagram of LedgerX

Data Flow Diagram

The Data Flow Diagram represents how information moves through the main functional stages of LedgerX. The user initiates wallet management operations, transaction information is maintained as transaction history, anomaly detection analyses the monitored activity, and the swap and recovery process uses the resulting security decision.

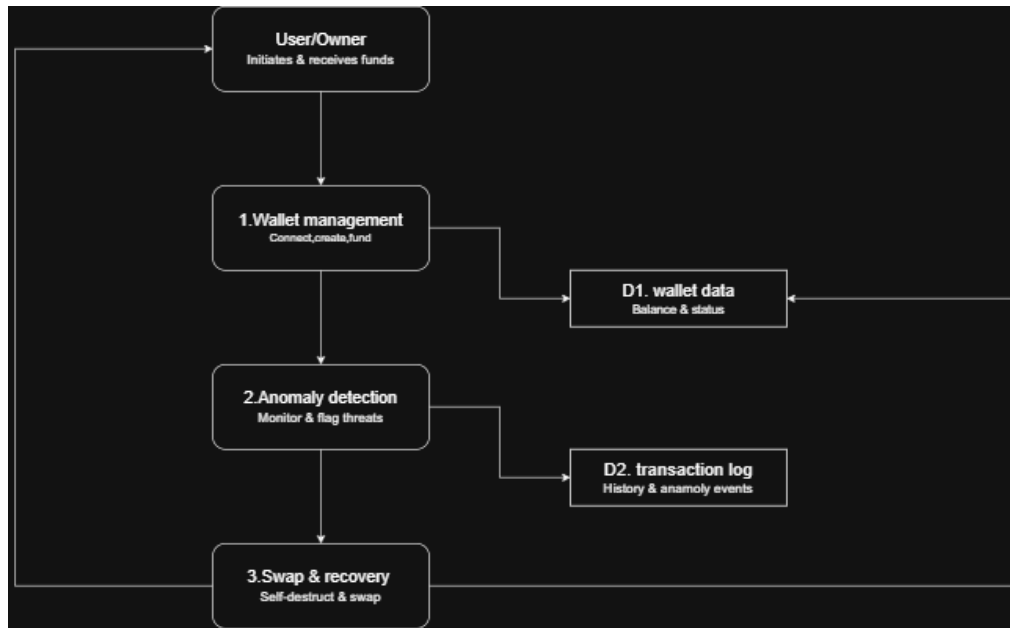


Figure 2.5: Data Flow Diagram of LedgerX

Class Diagram

The Class Diagram represents the structural relationships between the main logical components of the LedgerX smart-contract system. The diagram includes the Owner, Wallet, AnomalyGuard, and LiquidityPool components and illustrates their responsibilities and relationships.

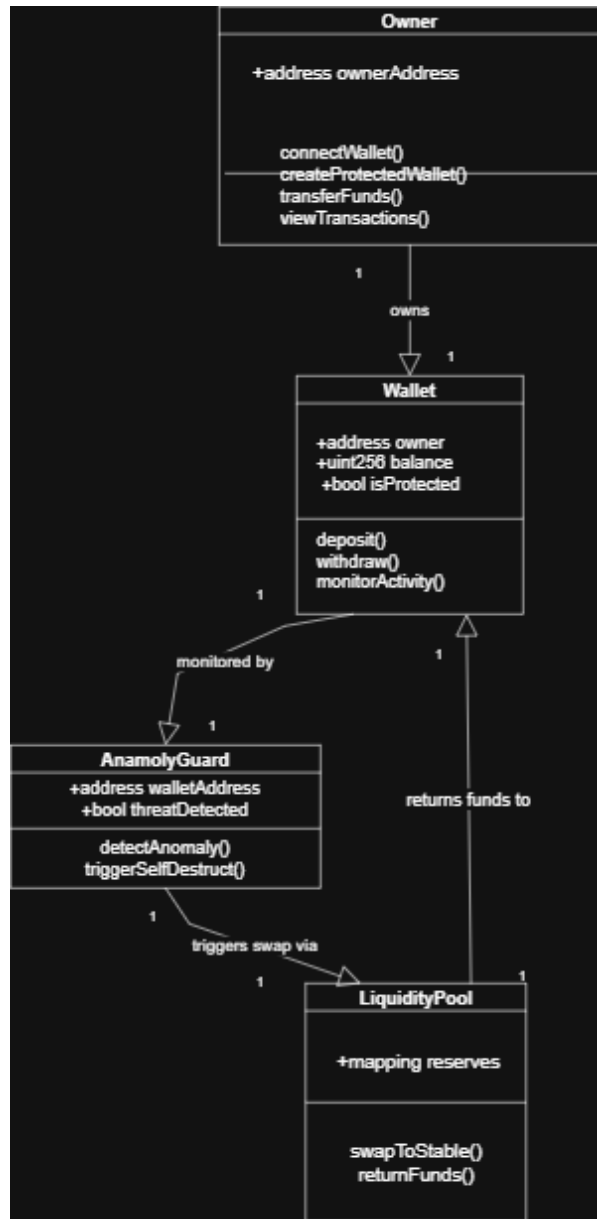


Figure 2.6: Class Diagram of LedgerX

2.1.2 Technical Stack

The LedgerX implementation combines web development, blockchain technology, smart contracts, Web3 integration, and machine-learning based anomaly detection.

- **Frontend:** Next.js and React are used to implement the web-based user interface.

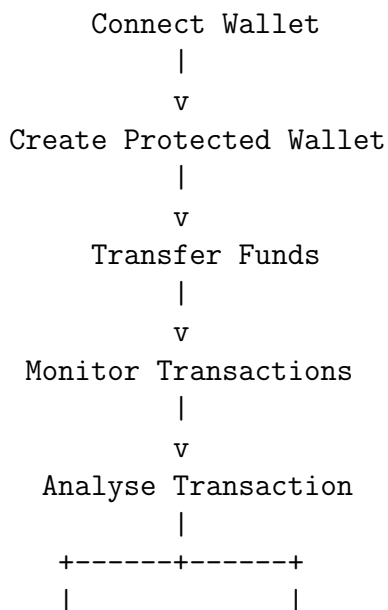
- **Programming Languages:** TypeScript/JavaScript are used for the web application and Solidity is used for blockchain smart contracts.
- **Blockchain and Smart Contracts:** Ethereum-compatible blockchain infrastructure is used for blockchain-based asset and security operations.
- **Smart Contract Development:** Hardhat and associated Ethereum development tools are used during smart-contract development.
- **Web3 Integration:** Web3 libraries are used to connect the frontend application with cryptocurrency wallets and blockchain contracts.
- **Machine Learning:** An Isolation Forest based approach is used for transaction anomaly detection.
- **Version Control:** Git and GitHub are used for source-code management and project collaboration.

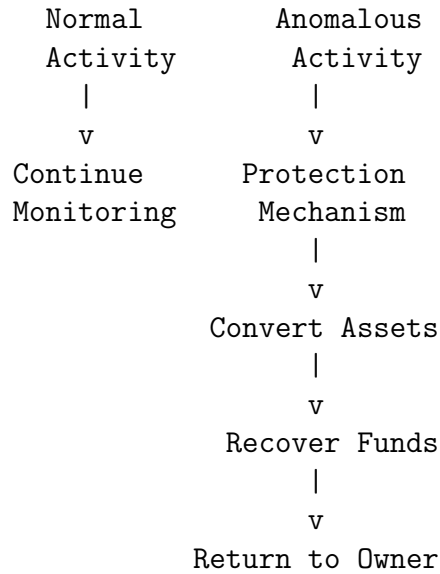
2.2 Implementation Details

2.2.1 Module 1: Core Functionality

The core functionality of LedgerX is the combination of protected wallet management, transaction monitoring, anomaly detection, and an automated asset-protection mechanism.

The main workflow of the module is:





Protected Wallet

The protected wallet is the primary security component of LedgerX. It provides an environment in which users can store and manage their digital assets while enabling the protocol to monitor relevant activity.

A user first connects a cryptocurrency wallet to the LedgerX application. The user can then create a protected wallet through the deployed smart-contract functionality. Assets can subsequently be transferred into the protected environment.

Transaction Monitoring

LedgerX provides transaction monitoring functionality through its web interface. The monitoring mechanism allows users to inspect activities associated with the protected wallet.

Transaction information is important for the security layer because the anomaly-detection component uses transaction characteristics to determine whether the observed activity is consistent with expected behaviour.

The monitoring functionality also improves transparency by allowing users to inspect the history of operations performed within the protocol.

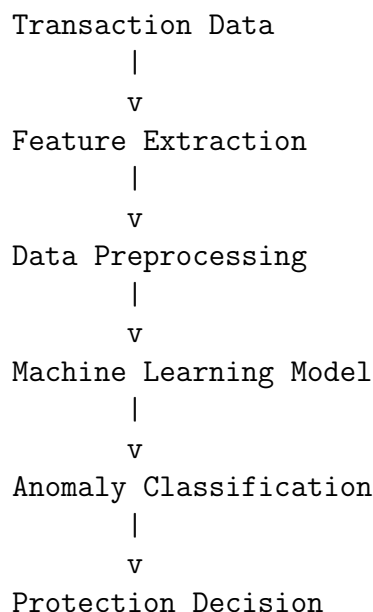
Anomaly Detection

The anomaly-detection component is responsible for identifying potentially suspicious transaction behaviour.

Transaction data is processed into relevant features that can be provided to the machine-learning model. The model evaluates the transaction and determines

whether its behaviour is sufficiently different from the expected transaction pattern to be classified as an anomaly.

The general processing pipeline is:



The project uses an Isolation Forest-based approach for anomaly detection. Isolation Forest is an unsupervised machine-learning algorithm designed to identify observations that differ from the majority of a dataset.

This approach is suitable for transaction monitoring because malicious or abnormal transactions may exhibit characteristics that differ significantly from normal transaction behaviour.

Automated Protection Mechanism

When suspicious activity is identified, LedgerX provides an automated protection workflow. The purpose of this mechanism is to reduce the time between detection of a potential threat and execution of a protective action.

The protection mechanism is implemented through smart contracts so that security-sensitive operations can be executed according to predefined blockchain rules.

The protection workflow involves detecting abnormal activity, initiating the protective mechanism, converting supported assets through the liquidity mechanism, and recovering the protected funds for the owner.

Liquidity Conversion and Fund Recovery

The protection workflow includes a liquidity conversion mechanism. When the protection mechanism is activated, assets can be converted into a safer or more stable

form through the liquidity component.

The purpose of this step is to reduce exposure of the user's assets during a potentially malicious event and provide a mechanism through which protected funds can subsequently be recovered.

After the conversion process, the resulting assets are intended to be returned to the designated owner through the smart-contract workflow.

Frontend Implementation

The current LedgerX frontend provides the primary user-facing interface for interacting with the protocol. The landing page introduces the protection concept and provides access to the wallet connection functionality. The interface also exposes navigation for sending assets, viewing transactions, managing tokens, creating assets, and accessing the chart/monitoring functionality.

Figure 2.7 shows the LedgerX landing page and wallet-connection interface.

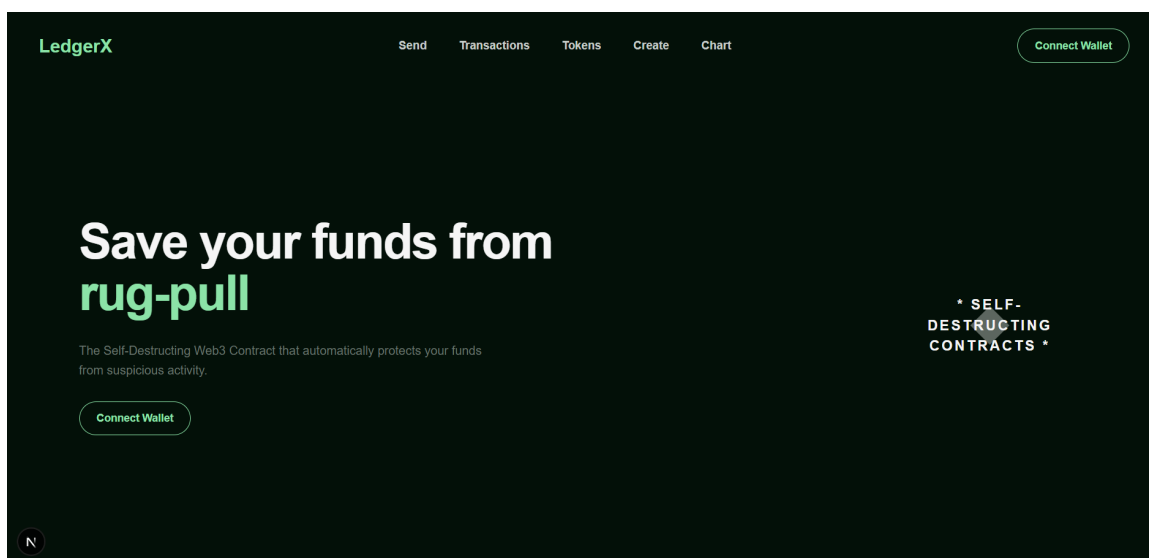


Figure 2.7: LedgerX Frontend Landing Page

Figure 2.8 shows the feature section of the frontend, highlighting real-time anomaly detection, the self-destruct mechanism, liquidity conversion, and fund recovery.

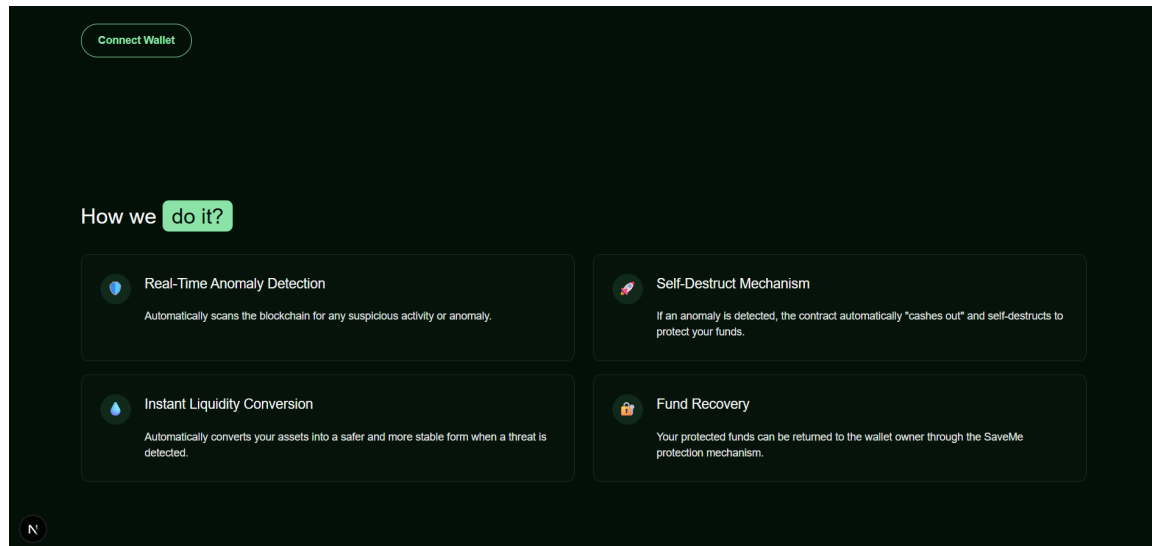


Figure 2.8: LedgerX Core Security Features

The frontend screenshots demonstrate the current implemented user interface and provide visual evidence of the core functionality presented in this stage of the project.

User Interaction Flow

The complete user interaction with the core functionality can be summarized as follows:

1. The user opens the LedgerX web application.
2. The user connects a supported cryptocurrency wallet.
3. The user creates a protected wallet.
4. The user transfers supported assets into the protected environment.
5. Transactions associated with the wallet are monitored.
6. Transaction characteristics are analysed by the anomaly detection component.
7. If the transaction is considered normal, the transaction continues through the normal workflow.
8. If potentially malicious activity is detected, the protection mechanism is activated.

9. The protection mechanism can initiate asset conversion and fund recovery through the smart-contract and liquidity components.
10. The protected funds are returned to the designated owner.
11. The user can inspect the resulting transaction history.

This completes the documentation of the core functionality included in the current stage of LedgerX.